CSX3007 (542) Quiz II 2/2025

This is a 3-hour online quiz (**14:00-17:00**), and students are not allowed to use any AI platforms, such as ChatGPT, Gemini, etc. to find answers (answers/formulas/equations obtained from such platforms are not valid for this exam). Use a scientific calculator instead of a calculator app. This is a **closed-book** examination (it carries a total of **70 points for 17 questions**)

* Indicates required question

1. Write your **name** , **ID**, and **section** in the below space: *

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1. (8 points) Consider the following **high-level language code** running on a MIPS system. **Show its MIPS assembly code translation, including complete memory operations with the data memory (DM)** for the variable allocation. Each variable's integer value is 32 bits. The **32-bit memory address** of the first byte of the variable***, a,*** is **0xAB108004.** The variables ***b*** and ***c*** use the consecutive byte addresses for their values in **DM**. Use the 32-bit registers **R1**, **R2**, and **R3** for ***a, b***, and ***c*** in memory and arithmetic operations, and register **R0 as the base register**, initialized to **0xAB102001**. *

MIPS instructions for the high-level code:

1. LW R1, 0x6003(R0) ; R1 = 0x00000019 (25)
2. ADDI R2, R2, 0x000F ; R2 = 0x0000000F (15)
3. SUB R2, R2, R1 ; R2 = 0xFFFFFFF6 (-10)
4. SW R2, 0x6007(R0)
5. ADDI R3, R2, 0x000A ; R3 = 0x00000014 (20)
6. SW R3, 0x600B(R0)

Pic1

```
int a = 25;
int b = 15 - a;
int c = b + 30;
```

The DM is utilized for variable storage:

32-bit Address	Data Byte	
0xAB108004	0x00	int a (32-bit) = 0x00000019 (25)
0xAB108005	0x00	
0xAB108006	0x00	
0xAB108007	0x19	
0xAB108008	0xFF	int b (32-bit) = 0xFFFFFFF6 (-10)
0xAB108009	0xFF	
0xAB10800A	0xFF	
0xAB10800B	0xF6	
0xAB10800C	0x00	int c (32-bit) = 0x00000014 (20)
0xAB10800D	0x00	
0xAB10800E	0x00	
0xAB10800F	0x14	

2. (2 points) Consider the MIPS system executing the following instruction as part of a benchmark test: **LW R1, 0x0006(R0)**. Assume the register **R0** is initialized with **0x00000006**. A section of the MIPS's byte-addressable **data memory** (**DM**) is shown in the **Figure below**. Find out the value of **R1** after the execution of the **LW** instruction. *

R1 = 0x5B4C6732

Pic2

Memory Address	Memory Word
0x00000008	0x23
0x00000009	0x5A
0x0000000A	0x1E
0x0000000B	0xA0
0x0000000C	0x5B
0x0000000D	0x4C
0x0000000E	0x67
0x0000000F	0x32
0x00000010	0xAF

3. (6 points) Below is a section of the MIPS pipeline system's **instruction memory (IM) is shown**. Based on that: (a). (2 points) What is the **Program Counter** (PC) value if the **IF stage** selects the instruction **0x41FA8C1C**? (b). (2 points) Show the **previous instruction** selected by the IF stage. (c). (2 points) Give the value of its **PC value** (from (**b**)). *

(a) (2 points) PC = 0x00000008 (b). (2 points) 0x3B1ED031 (c). (2 points) PC= 0x00000004

Pic3

Memory Address	Memory Word
0x00000000	0x8A
0x00000001	0xF3
0x00000002	0x67
0x00000003	0x52
0x00000004	0x3B
0x00000005	0x1E
0x00000006	0xD0
0x00000007	0x31
0x00000008	0x41
0x00000009	0xFA
0x0000000A	0x8C
0x0000000B	0x1C

4. (3 points) Answer the question that is shown in the figure below: *

There are 32 registers in the RF. That's why 5-bit address is needed to access the RF. The ID and WB stages are dependent on the RF.

Pic4

Why does the MIPS system use a **5-bit address** to access its register file (RF)? Which pipeline stages depend on the RF?

5. (2 points) Answer the question that is shown in the figure below: *

Speedup = no. of stages/(Ideal CPI + stall (delay)) = 5/1+0.78 = 2.81 times

Pic5

Assume that the MIPS pipelined system has imposed a stall of **0.78** clock cycles per instruction for a particular user process. What is the **system's speedup**?

6. (4 points) Answer the question that is shown in the figure below: *

Average instruction Exe-time (unpipeline):
= 3 ns x [(450x4 + 200x5 + 350x6)/(450+200+350)] = 3 ns x 4.9 = 14.7 ns

Average instruction Exe-time (pipeline):
= (3 ns + 0.25 ns) x pipeline CPI (ideal) = 3.25 ns x 1 = 3.25 ns

Speedup from pipeline = 14.7ns/3.25ns = 4.523 times

Pic6

Consider an **unpipelined** processor. It has a **3 ns** clock cycle and uses **4 cycles for ALU operations, 5 for branches, and 6** for memory operations. Assume that the relative instruction frequencies of these operations are **450**, **200**, and **350**, respectively. **Pipelining** the processor adds **0.25 ns** of overhead to the clock (ignore other pipeline latencies and hazards). Find out how much speedup the instruction execution rate will gain from the pipeline.

7. (5 points) Consider the given MIPS instruction sequence: (a). (2 points) Show the data-dependency causing registers and their corresponding instructions (using line numbers). (b). (3 points) show the **clock cycles** and **stages** of forwarding's and stalls caused by data dependency instructions. *

Forwarding

	1	2	3	4	5	6	7	8	9	10	11	12
1. ADDI R1, R4, 0x0008	IF	ID	EX	MEM	WB							
2. ADDI R2, R1, 0x0007		IF	ID	EX	MEM	WB						
3. SUB R4, R2, R1			IF	ID	EX	MEM	WB					
4. LW R5, 5(R6)				IF	ID	EX	MEM	WB				
5. SUB R6, R5, R1					IF	ID	STALL	EX	MEM	WB		
6. AND R7, R1, R3						IF	STALL	ID	EX	MEM	WB	
7. SUBI R8, R7, 0x0005							STALL	IF	ID	EX	MEM	WB

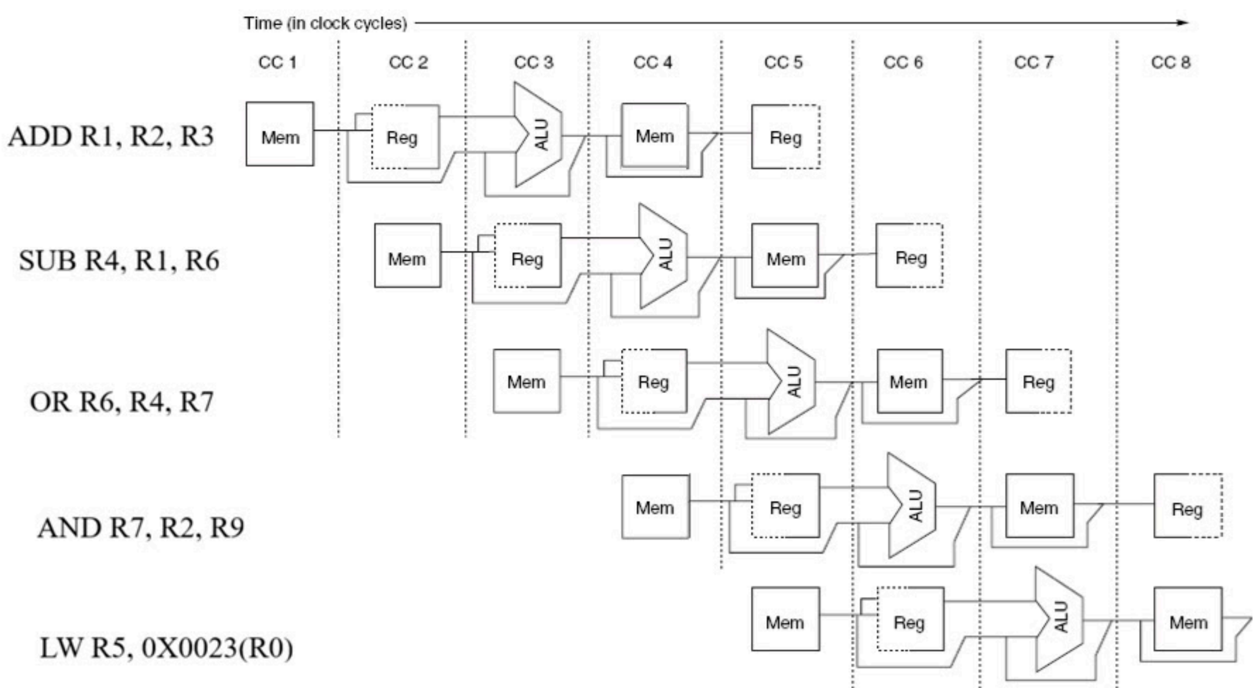
Pic7

```
1. ADDI R1, R4, 0x0008
2. ADDI R2, R1, 0x0007
3. SUB R4, R2, R1
4. LW R5, 5(R6)
5. SUB R6, R5, R1
6. AND R7, R1, R3
7. SUBI R8, R7, 0x0005
```

8. (2 points) Consider the pipeline data path of the MIPS system shown in **Figure below**. Is there any structural hazard in the system? If yes, describe the possible ways (if any) to eliminate it from the system. *

There is no Structural Hazard in the system (reason: the first instruction is not a LW instruction, see the figure!)

Pic8



9. (3 points) Based on the following MIPS instruction sequence, determine the **values** (in hexadecimal) of the registers **R1, R2,** and **R3** after the instruction execution (assume that the register **R0** is initialized with **0x00000002**). *

R1 = 0x0000000C
R2 = 0x0000000B
R3 = 0x0000000D

Pic9

```
1. ADDI R1, R0, 0X0004
2. ADDI R2, R1, 0X0005
3. SLL R1, R1, 0X0001
4. BEQ R1, R2, TARGET
5. ADD R3, R2, R1
   TARGET:
6. SUBI R3, R3, 0X000A
```

10. (2 points) Assume that a high-level language compiler is running on an MIPS system and it generates the following assembly code (where R0, R1, R2, R3, and R4 are 32-bit general purpose registers and are loaded with integer variables *f*, *g*, *h*, *i* and *j*, respectively). Show its source code. *

if (i == j)
f = g + h ;
else
f = f - i;

Pic10

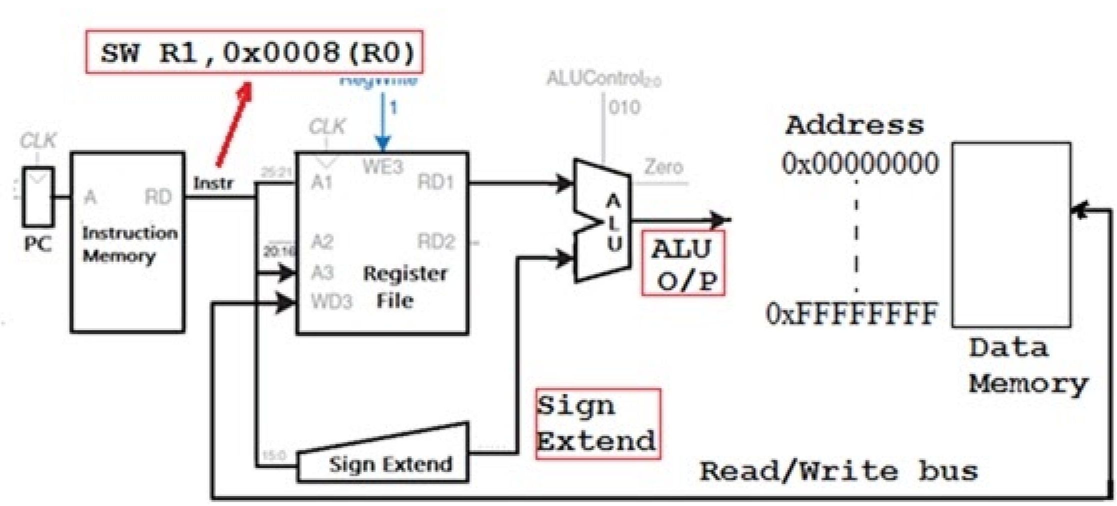
```
     BNE R3, R4, Next
      ADD R0, R1, R2
      J Exit
Next:
      SUB R0, R0, R3
Exit:
```

11. (7 points) Answer the following questions based on the single-cycle data path of the MIPS system shown in the **Figure** below. Assume that the registers **R0** and **R1** have initialized with **0x00000007** and **0xC123FAE7**: (a). (1 point) Show the **I/P of the Sign Extend unit** (in hexadecimal). (b). (1 point) Show the **O/P of the Sign Extend unit** (in hexadecimal). (c). (1 point) Show the **O/P of the ALU** (in hexadecimal). (d). (4 points) Show the result of the **SW** operation in **Data Memory** (show each byte word with its complete byte address).

(a). (1 point) 0x0008 (b). (1 point) 0x00000008
(c). (1 point) 0x0000000F

(d) (4 points)

DM address: Byte
0x0000000F: 0xC1
0x00000010: 0x23
0x00000011: 0xFA
0x00000012 : 0xE7



12. (2 points) Answer the question that is given in the figure below: *

DM (Data Memory)

Pic12

One of the **four state elements** of the MIPS system support the MEM stage of the MIPS's pipeline. Which is that?

13. (6 points) Answer the following based on the MIPS instruction format given in the **Figure** below: (a). (1 point) What is the instruction type (in terms of the operands)? (b). (1 point) Show an example instruction (in assembly code). (c). (1 point) How many opcodes are possible? (d). (1 point) How many registers are in the system? (e). (1 point) How many functions are in the system? (d). (1 point) What is the size of the CPU? *

(a). R-type (b). ADD R1, R2, R3 (c). 64 (d). 32 (e). 64 (f). 32-bit

Pic13

op	rs	rt	rd	shamt	funct
6 bits	5 bits	5 bits	5 bits	5 bits	6 bits

14. (6 points) A **0.25 ns** clock cycled unpipelined processor executing a benchmark program with the instruction mix and clock cycle count that is shown below. Determine its **CPI**, **CPU time (**execution time), and **MIPS rate**. *

CPI = (0.40 x 3) + (0.30 x 2) + (0.20 x 4) + (0.10 x 5) = 3.1

CPU time = IC x CPI x Clock_cycle time
= 100 x 3.1 x 0.25 x 10^{-9} = 77.5 ns

MIPS rate = IC/(CPU time x 10^6) = 1/(3.1 x 0.25 x 10^{-9}x10^6)
= 1000/(3.1 x 0.25) = 1290

Pic14

Instruction	Instruction Count	Clock cycle count
Integer arithmetic	40%	3
Data transfer	30%	2
Floating point	20%	4
Control transfer	10%	5

15. (4 points) Answer the question that is given in the figure below: *

Speedup = 1/((1-a) + a/n)

DP4600: a = 0.8, n =8; Speedup = 1/((1-0.8) + 0.8/8) = 3.33 times

H6600: a = 0.7, n = 8; Speedup = 1/((1-0.7) + 0.7/8) = 2.58 times

DP4600 is more efficient!

Pic15

Suppose that you must choose between DP4600 (a **8**-processor system) and H6600 (an **8**-processor system) servers. Both are clocked at similar clock rates. Assume that scheduling a task on the H6600 server incurs a 30% sequential overhead, whereas on the DP4600 it incurs only a 20% overhead. Which system is more efficient for running your task (show your calculation steps)?

16. (4 points) The execution rates (in jobs per hour) of various scientific programs on two computers (A and B) are shown in the following table. Based on the rate data, which one has the better central tendency? *

Comp A: 4/(1/0.35 + 1/0.28 +1/0.43+1/0.56) = 0.3794
Comp B: 4/(1/0.38 + 1/0.26 + 1/0.39 + 1/0.49) = 0.361

It seems that Comp A has slightly higher central tendency than Comp B.

Pic16

Comp A	Comp B
0.35	0.38
0.28	0.26
0.43	0.39
0.56	0.49

17. (4 points) Answer the question that is shown in the figure below: *

Weight of J1 = 45/195 = 0.2308, Weight of J2 = 55/195 = 0.2821
weight of J3 = 60/195 = 0.3077, Weight of J4 = 35/195 = 0.1795
Weighted Arithmetic Mean:
(0.64 x 0.2308 + 0.45 x 0.2821 + 0.53 x 0.3077 + 0.73 x 0.1795) =
0.147712 + 0.12695 + 0.16308 + 0.131035 = 0.5688 ns

Pic17

Consider **four jobs** being run on a corporate computer. The observed optimal execution times in nano-seconds were **0.64**, **0.45**, **0.53**, and **0.73**. To obtain these optimal time values, the jobs must be executed many times: **Job1** was executed 45 times, **Job2** was executed 55 times, **Job3** was executed 60 times, and **Job4** was executed 35 times. What is the *central tendency* of the system?